

Eco-Cart – Eco-Friendly Online Shopping Platform

Requirement Analysis Document

Introduction:

Eco-Cart is an environmentally friendly e-commerce platform that promotes sustainable living by offering only eco-friendly, organic, biodegradable, reusable, and recyclable products. The platform connects environmentally conscious customers with trusted sellers who provide sustainable alternatives to everyday products. Eco-Cart aims to reduce plastic waste, encourage green shopping, and support eco-friendly businesses while providing a secure and convenient online shopping experience.

Objectives:

- Promote sustainable and environmentally friendly shopping.
- Sell only certified organic, biodegradable, recyclable, and reusable products.
- Reduce the use of plastic through eco-friendly packaging.
- Support local organic farmers and sustainable brands.
- Provide secure online shopping and payment services.
- Educate customers about eco-friendly living.
- Encourage customers to make environmentally responsible purchasing decisions.

Major Activities in the System:

A. Customer Registration & Authentication:

- User registration using email or mobile number.
- Secure login with encrypted passwords.
- OTP/email verification.
- Profile management.
- Manage delivery addresses.
- View order history.

B. Eco-Friendly Product Browsing:

Customers can browse products under categories such as:

- Organic Food
- Natural Personal Care
- Bamboo Products
- Reusable Kitchen Items
- Eco-friendly Stationery

- Cloth Bags
- Recycled Home Decor
- Sustainable Fashion
- Organic Farming Products

Each product displays:

- Product images
- Description
- Organic certification
- Eco-score
- Material used
- Environmental benefits
- Customer ratings and reviews

C. Product Search & Filter:

Customers can search products using:

- Product name
- Category
- Brand
- Price
- Organic Certified
- Plastic-Free
- Recyclable
- Compostable
- Vegan
- Eco Rating

D. Shopping Cart Management:

Customers can:

- Add products to cart
- Remove products
- Update quantity
- Save for later

- View total amount
- View estimated carbon footprint saved
- Apply eco-discount coupons

E. Order Placement:

- Checkout process
- Select shipping address
- Choose eco-friendly packaging
- Apply coupons
- Confirm order
- Generate invoice
- View order history
- Cancel or return eligible products

F. Payment Processing:

Supported payment methods include:

- UPI
- Credit Card
- Debit Card
- Net Banking
- Digital Wallets
- Cash on Delivery

Features:

- Secure encrypted transactions
- Payment confirmation
- Digital receipt generation

G. Sustainable Delivery & Shipment Tracking:

- Real-time shipment tracking
- Eco-friendly delivery partners
- Carbon-efficient delivery routes
- Order status updates:
 - Order Confirmed

- Packed
- Shipped
- Out for Delivery
- Delivered
- SMS and Email notifications

H. Admin Management:

Administrator can:

- Manage eco-friendly product listings
- Verify organic certifications
- Approve sustainable sellers
- Monitor customer orders
- Manage inventory
- View sales reports
- Handle customer support
- Monitor carbon savings statistics

Functional Requirements:

Customer Management	Registration, Login, Profile Management, Address Management
Eco Product Catalog	Add/View Organic Products, Search, Filter, Eco-score Display
Shopping Cart	Add/Remove Products, Update Quantity, Save for Later
Wishlist	Add/Remove Wishlist Items
Order Management	Place Orders, Cancel Orders, Return Products, Generate Invoice
Payment Gateway	Secure Payment Processing, Receipt Generation
Shipment Tracking	Live Order Tracking, Delivery Notifications
Sustainability Dashboard	Display Carbon Savings, Plastic Waste Reduction, Eco Impact
Seller Management	Verify Organic Sellers, Manage Certifications
Admin Panel	Product Management, Customer Management, Reports, Inventory Management

Non-Functional Requirements:

Performance

- Fast product search and checkout.
- Support thousands of concurrent users.

Security

- Encrypted passwords.
- Secure payment gateway.
- Customer data privacy.
- Role-based access control.

Reliability

- High system availability.
- Accurate order processing.
- Automatic data backup.

Scalability

- Support growing numbers of products and customers.
- Easily integrate new eco-friendly sellers.

Usability

- User-friendly interface.
- Mobile responsive design.
- Easy navigation for all age groups.

Sustainability

- Encourage paperless invoices.
- Promote eco-friendly packaging.
- Display environmental impact for every purchase.
- Minimize carbon footprint in logistics.

Features of Eco-Cart:

- Only certified organic and eco-friendly products.
- Recyclable and biodegradable product categories.
- Eco-score displayed for every product.
- Carbon footprint savings tracker.

- Plastic-free packaging option.
- Green delivery partners.
- Support for local organic farmers and sustainable businesses.
- Sustainability dashboard showing the customer's positive environmental impact.
- Reward points ("Green Points") for eco-friendly purchases.
- Educational blogs and tips on sustainable living.

Conclusion:

Eco-Cart is more than just an online shopping platform—it is a sustainable marketplace that promotes green living and environmental responsibility. By offering only organic, recyclable, biodegradable, and eco-friendly products, Eco-Cart empowers customers to make environmentally conscious purchasing decisions. The platform combines secure e-commerce features with sustainability-focused innovations, creating a convenient shopping experience while contributing to a cleaner and healthier planet.